Loan Default Prediction Using Machine Learning

Machine Learning: Classification
Course: Engineering Project (Machine Learning: Classification)
Project Topic: Loan Default Prediction
Dataset: Give Me Some Credit, Kaggle
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Abstract

Loan default prediction is an important problem in financial risk management because inaccurate lending decisions can lead to major financial losses for banks and lending institutions. This project applies machine learning techniques to predict whether a borrower will default on a loan using the “Give Me Some Credit” dataset from Kaggle. The project follows a complete machine learning workflow, including exploratory data analysis, preprocessing, missing value handling, class imbalance analysis, baseline modeling, advanced modeling, hyperparameter tuning, error analysis, interpretability analysis, and risk scoring.

Several classification models were implemented and compared, including Logistic Regression, Balanced Logistic Regression, Random Forest, and Tuned Random Forest. Since the dataset was highly imbalanced, model evaluation focused not only on accuracy but also on precision, recall, F1-score, and ROC-AUC. RandomizedSearchCV with cross-validation was used for hyperparameter tuning. In addition, feature importance and SHAP analysis were applied to interpret the model predictions and identify the strongest predictors of loan default risk.

The Random Forest models achieved the strongest overall performance, with ROC-AUC scores of approximately 0.864. The most influential predictors were revolving credit utilization and delinquency-related features, especially previous late payment behavior. A risk scoring system was also developed to classify borrowers into Low Risk, Medium Risk, and High Risk groups based on predicted default probabilities. The project demonstrates how machine learning can support financial risk assessment while also highlighting important ethical concerns, including bias, responsible AI use, and the limitations of automated financial decision-making.

1. Introduction

Loan default prediction is a major application of machine learning in the financial sector. A loan default occurs when a borrower fails to repay their debt according to the agreed repayment schedule. For financial institutions, default prediction is important because poor lending decisions may increase financial losses, reduce profitability, and weaken the stability of lending systems. For borrowers, fair and accurate credit assessment is also important because lending decisions can directly affect access to financial opportunities.

Traditional credit scoring systems have often relied on statistical models and manually designed rules to classify borrowers into different risk groups. Credit scoring has long been used to separate applicants into relatively “good” and “bad” risk categories based on financial and behavioral indicators (Hand & Henley, 1997). However, modern financial datasets can contain complex nonlinear patterns that may be difficult to capture using simple rule-based systems alone. Machine learning methods can help identify these patterns by learning from historical borrower data.

This project focuses on predicting loan default risk using the “Give Me Some Credit” dataset from Kaggle. The task is formulated as a binary classification problem. The target variable indicates whether a borrower experienced serious delinquency within a two-year period. The goal is to build and evaluate machine learning models that can classify borrowers as default or non-default cases based on financial and demographic features.

The project uses a complete machine learning pipeline. First, exploratory data analysis is performed to understand the dataset, identify missing values, detect class imbalance, and examine feature distributions. Next, preprocessing techniques are applied, including removal of unnecessary columns, handling invalid values, median imputation for missing numerical values, and train-test splitting. Several classification models are then trained, including Logistic Regression, Balanced Logistic Regression, Random Forest, and Tuned Random Forest. The models are evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices.

Because financial default datasets are often imbalanced, accuracy alone is not sufficient for evaluating model performance. A model can achieve high accuracy by predicting the majority class most of the time while failing to detect actual defaulters. Therefore, this project places special emphasis on recall, F1-score, ROC-AUC, and the business meaning of false positives and false negatives. The project also includes interpretability analysis using Random Forest feature importance and SHAP values, which help explain how borrower characteristics influence model predictions.

The main objectives of this project are:

- to analyze the loan default dataset using exploratory data analysis,
- to prepare the dataset for machine learning without data leakage,
- to implement baseline and advanced classification models,
- to evaluate model performance using appropriate metrics for imbalanced classification,
- to tune model hyperparameters using cross-validation,
- to interpret model predictions using feature importance and SHAP analysis,
- and to develop a practical risk scoring system for borrower classification.

Overall, this project demonstrates how machine learning can be used as a decision-support tool for financial risk assessment while recognizing the need for responsible use, fairness, and human oversight.

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2. Related Work

Credit scoring and loan default prediction have been studied for many years in both statistics and machine learning. Hand and Henley (1997) described credit scoring as a statistical classification problem where applicants are classified into risk groups based on observed characteristics. Their review emphasized that many statistical classification techniques can be applied to consumer credit scoring, but also noted that commercial credit data is often difficult to access publicly due to confidentiality concerns.

Logistic Regression has historically been one of the most common models used in credit scoring because it is simple, interpretable, and suitable for binary classification problems. Its coefficients can be examined to understand the direction of relationships between predictors and the target variable. However, Logistic Regression may struggle when relationships between variables and default risk are nonlinear or involve complex interactions.

Tree-based machine learning models, including Random Forest, have become popular alternatives for credit risk prediction. Random Forest is an ensemble learning method that combines multiple decision trees to improve prediction accuracy and reduce overfitting (Breiman, 2001). This makes it useful for tabular datasets where feature interactions and nonlinear patterns may exist. Ensemble methods have also been shown to perform strongly in credit scoring benchmarks. Lessmann et al. (2015) compared multiple classification algorithms for credit scoring and found that modern machine learning methods, especially ensemble-based approaches, can achieve strong predictive performance.

Another important challenge in loan default prediction is class imbalance. In many credit datasets, most borrowers do not default, while only a smaller proportion experience serious delinquency. This imbalance can cause models to favor the majority class and perform poorly on the minority class. Because of this, evaluation metrics such as recall, F1-score, ROC-AUC, and confusion matrices are important for understanding whether a model is actually detecting default cases rather than only predicting non-default cases.

Model interpretability is also important in financial machine learning. Lending decisions affect individuals and may have ethical, legal, and social consequences. Therefore, financial models should not only perform well but should also be explainable. SHAP, introduced by Lundberg and Lee (2017), provides a unified framework for interpreting model predictions by assigning each feature a contribution value. In this project, SHAP analysis is used to better understand how features such as credit utilization, late payments, age, debt ratio, and income influence predicted default risk.

This project builds on these ideas by combining traditional baseline models, ensemble-based models, hyperparameter tuning, class imbalance handling, and interpretability methods in one complete loan default prediction workflow.

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3. Dataset Description

The dataset used in this project is the "Give Me Some Credit" dataset from Kaggle. The purpose of the dataset is to support credit scoring and financial distress prediction. The prediction task is to estimate whether a borrower will experience serious delinquency within two years.

The original training dataset contains 150,000 borrower records. The target variable is `SeriousDlqin2yrs`, where:

- `0` indicates that the borrower did not default,
- `1` indicates that the borrower defaulted or experienced serious delinquency.

The dataset includes numerical features related to borrower financial behavior and demographic information. Important features include:

Feature	Description
RevolvingUtilizationOfUnsecuredLines	Ratio of unsecured revolving credit utilization
age	Borrower age
NumberOfTime30-59DaysPastDueNotWorse	Number of times borrower was 30–59 days past due
DebtRatio	Monthly debt payments divided by income or related financial ratio
MonthlyIncome	Borrower monthly income
NumberOfOpenCreditLinesAndLoans	Number of open credit lines and loans
NumberOfTimes90DaysLate	Number of times borrower was 90 or more days late
NumberRealEstateLoansOrLines	Number of mortgage or real estate loans
NumberOfTime60-89DaysPastDueNotWorse	Number of times borrower was 60–89 days past due
NumberOfDependents	Number of dependents

During initial inspection, the dataset contained an unnecessary `Unnamed: 0` column. This column represented an index and was removed because it did not provide meaningful information for prediction.

Missing values were identified in `MonthlyIncome` and `NumberOfDependents`. Because financial variables such as income may contain outliers, median imputation was selected as a robust method for filling missing numerical values.

The dataset also contained one record with an age value of zero. Since this is unrealistic for a loan applicant, that row was removed before model training. This cleaning step improved the validity of the dataset without meaningfully reducing the dataset size.

The target distribution showed strong class imbalance. Most borrowers belonged to the non-default class, while default cases represented a much smaller proportion of the dataset. This imbalance influenced both model design and evaluation strategy.

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4. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the dataset structure, detect data quality issues, analyze class distribution, and identify potentially important features.

The first major observation was the imbalance in the target variable. The majority of borrowers did not default, while only a small proportion were labeled as default cases. This imbalance is important because a model could achieve high accuracy by predicting most borrowers as non-default, while still failing to detect actual risky borrowers.

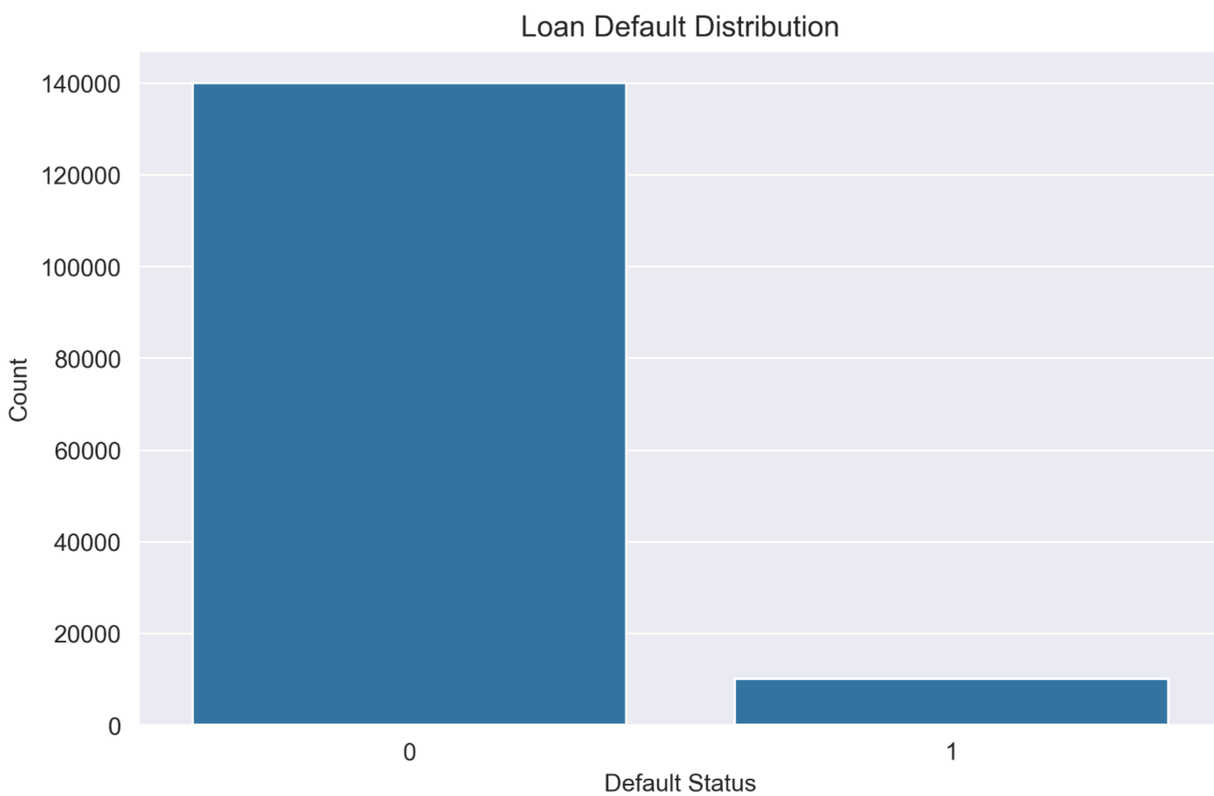

Figure 1. Loan default distribution.

Figure 1 shows the distribution of default and non-default cases. The non-default class is much larger than the default class. This confirms that accuracy alone is not enough to evaluate the model.

Missing value analysis showed that `MonthlyIncome` had a relatively high missing value percentage, while `NumberOfDependents` had a smaller missing value percentage. These missing values were handled using median imputation during preprocessing.

The age distribution showed that most borrowers were middle-aged, but the minimum value inspection revealed one invalid age value equal to zero. This record was removed because it was not realistic for a borrower.

Monthly income analysis showed a highly right-skewed distribution. The median income was much lower than the maximum income, indicating the presence of extreme outliers. For visualization purposes, income values were temporarily filtered to better understand the main distribution. However, the full dataset was retained for model training, with missing values handled through preprocessing.

A correlation heatmap was generated to examine relationships among numerical features.

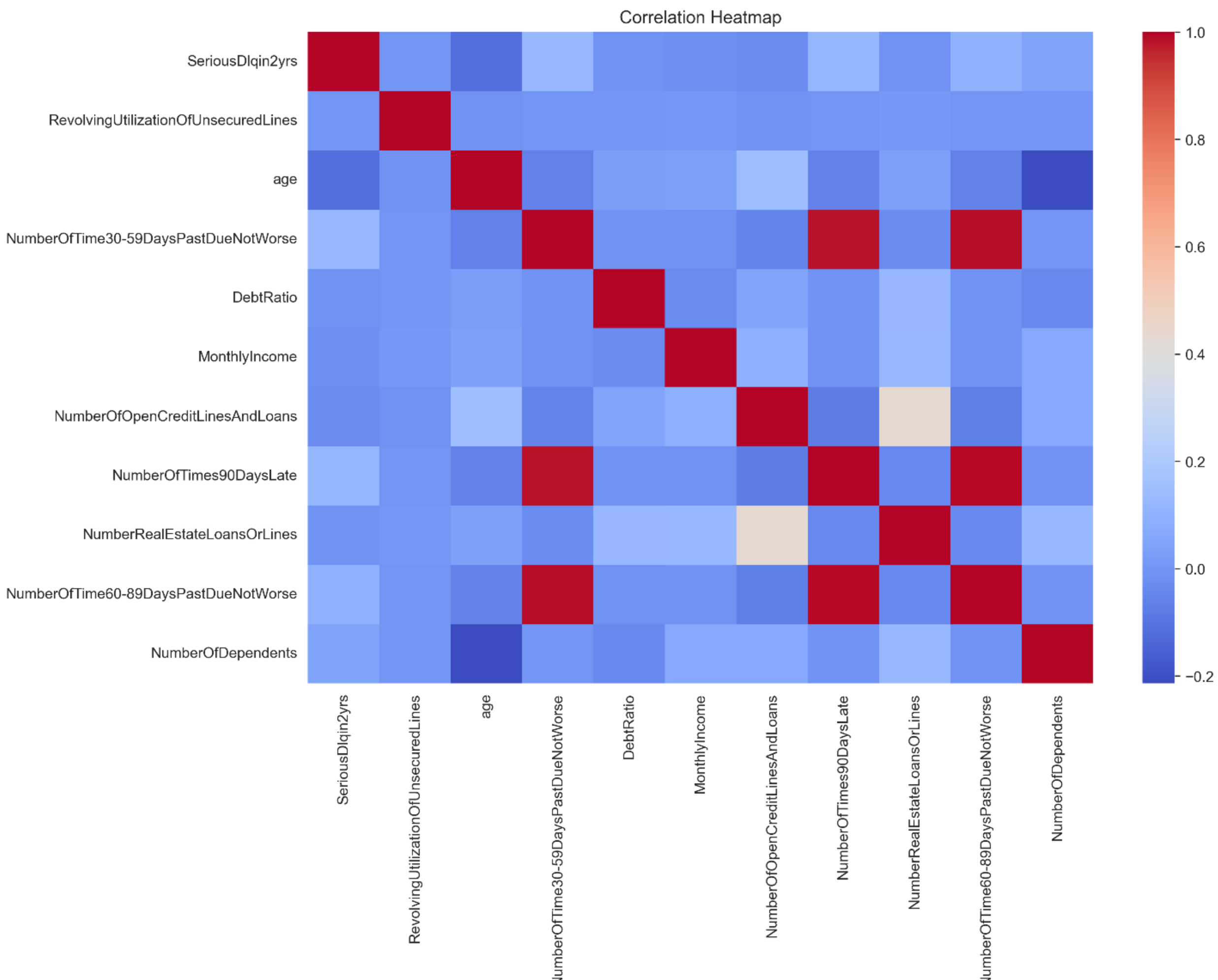

Figure 2. Correlation heatmap.

Figure 2 shows that delinquency-related variables are strongly related to each other. This is expected because borrowers who have been late in one payment category may also be more likely to have late payments in other categories. These features were later found to be important predictors of default risk.

Overall, EDA revealed four major insights:

1. the dataset is highly imbalanced,
2. some features contain missing values,
3. income-related features contain extreme values,
4. delinquency history and credit utilization are likely important predictors of loan default.

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5. Data Preprocessing and Feature Engineering

The preprocessing stage prepared the raw dataset for machine learning. The first step was removing the unnecessary `Unnamed: 0` column because it represented an index rather than a meaningful feature. The invalid age record where `age = 0` was also removed.

The dataset was then separated into input features (`X`) and the target variable (`y`). The target variable was `SeriousDlqin2yrs`, and all remaining relevant borrower features were used as predictors.

The data was split into training and testing sets using an 80/20 split. Stratified sampling was applied to preserve the original class distribution in both training and testing data. This is especially important for imbalanced datasets because random splitting without stratification may create unequal class distributions across train and test sets.

Missing values were handled using median imputation. Median imputation was selected because variables such as `MonthlyIncome` and `DebtRatio` can contain extreme outliers. The median is less affected by outliers than the mean, making it more suitable for this type of financial data.

For Logistic Regression models, StandardScaler was used to standardize numerical variables. Scaling helps Logistic Regression because it is sensitive to feature magnitude. Random Forest models do not require feature scaling because tree-based models split features based on threshold values rather than distance-based calculations.

The preprocessing was implemented using Scikit-learn pipelines. Pipelines are useful because they combine preprocessing and modeling steps into one object, reducing the risk of data leakage. Data leakage occurs when information from the test set influences training, leading to overly optimistic evaluation results.

Feature engineering in this project focused on using meaningful borrower behavior variables such as credit utilization, payment history, age, debt ratio, and income. These features are directly related to financial risk and align with the project objective of predicting loan default.

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6. Baseline Modeling

Baseline models were implemented to establish initial performance benchmarks. Logistic Regression was selected as the first baseline model because it is a common method for binary classification and has historically been used in credit scoring applications.

6.1 Logistic Regression

The standard Logistic Regression model used median imputation and feature scaling. The model achieved high accuracy, but its recall was very low. This means that although the model correctly predicted many non-default cases, it failed to detect most actual default cases.

This result demonstrates why accuracy alone is misleading for imbalanced classification problems. Since most borrowers did not default, the model could achieve high accuracy by mostly predicting the majority class.

6.2 Balanced Logistic Regression

To address class imbalance, a second Logistic Regression model was trained using class weighting. Class weighting gives more importance to the minority class during training. This significantly improved recall, meaning the model detected more default cases. However, this improvement came at the cost of lower precision and lower overall accuracy.

This trade-off is important in loan default prediction. Increasing recall helps identify more risky borrowers, but it may also increase false positives, meaning more non-default borrowers may be incorrectly classified as risky.

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7. Advanced Modeling

Random Forest was selected as the advanced model because it is well suited for tabular data and can capture nonlinear relationships between features. Random Forest combines multiple decision trees and averages their predictions to improve generalization and reduce overfitting (Breiman, 2001).

The Random Forest model was trained using class weighting to account for the imbalanced target variable. Compared with Logistic Regression models, Random Forest achieved better recall, F1-score, and ROC-AUC. This suggests that the model was better able to distinguish between default and non-default borrowers.

Random Forest also provides feature importance values, making it useful for interpretability. This is especially valuable in financial applications, where understanding why a borrower is considered risky is important for responsible decision-making.

8. Hyperparameter Tuning and Validation

Hyperparameter tuning was performed using RandomizedSearchCV. The goal was to improve the Random Forest model by testing different combinations of hyperparameters. Cross-validation was used during tuning to evaluate model performance more reliably across different data splits.

The hyperparameter search space included:

- number of trees (`n_estimators`),
- maximum tree depth (`max_depth`),
- minimum samples required to split a node (`min_samples_split`),
- minimum samples required at a leaf node (`min_samples_leaf`).

The tuning process used ROC-AUC as the scoring metric because ROC-AUC evaluates how well the model separates default and non-default cases across different thresholds.

In the final run, the best configuration was approximately:

Hyperparameter	Best Value
n_estimators	100
max_depth	10
min_samples_split	2
min_samples_leaf	4

The best cross-validated ROC-AUC score was approximately 0.857. This shows that the tuned model had strong validation performance.

Hyperparameter tuning demonstrated an important trade-off. The tuned Random Forest achieved similar ROC-AUC performance to the original Random Forest, but small changes occurred in precision, recall, and F1-score. This confirms that improving one metric may slightly reduce another, especially in imbalanced classification problems.

9. Model Evaluation and Error Analysis

The models were evaluated using Accuracy, Precision, Recall, F1-score, ROC-AUC, and confusion matrices. These metrics provide different perspectives on model performance.

- Accuracy measures the overall proportion of correct predictions.
- Precision measures how many predicted default cases were actually defaults.
- Recall measures how many actual default cases were successfully detected.
- F1-score balances precision and recall.
- ROC-AUC measures the model's ability to distinguish between classes across thresholds.

The final rounded performance table is shown below.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.934	0.593	0.044	0.083	0.712
Balanced Logistic Regression	0.770	0.178	0.673	0.281	0.800
Random Forest	0.819	0.230	0.729	0.350	0.864
Tuned Random Forest	0.821	0.232	0.725	0.351	0.864

The baseline Logistic Regression model achieved the highest accuracy, but its recall was extremely low. This means it failed to detect most default cases. The Balanced Logistic Regression model improved recall by applying class weighting, but precision and accuracy decreased.

The Random Forest models achieved the strongest overall performance. Both the original and tuned Random Forest models achieved ROC-AUC scores of approximately 0.864. The tuned Random Forest slightly improved some metrics, while the original Random Forest maintained slightly stronger recall in some runs. Overall, the Random Forest family provided the best balance between detecting default cases and maintaining overall classification quality.

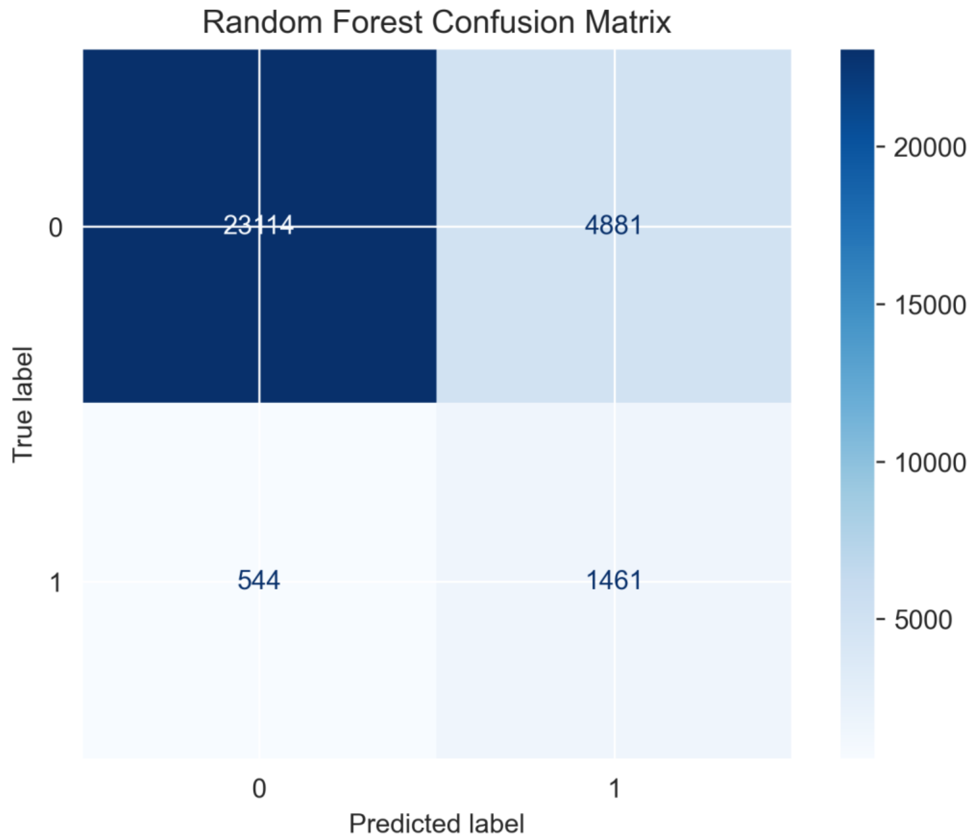

Figure 3. Random Forest confusion matrix.

Figure 3 shows the Random Forest confusion matrix. The matrix helps explain the model's correct and incorrect predictions. False negatives are especially important in loan default prediction because they represent borrowers who actually defaulted but were predicted as non-default. False positives are also important because they represent borrowers incorrectly classified as risky.

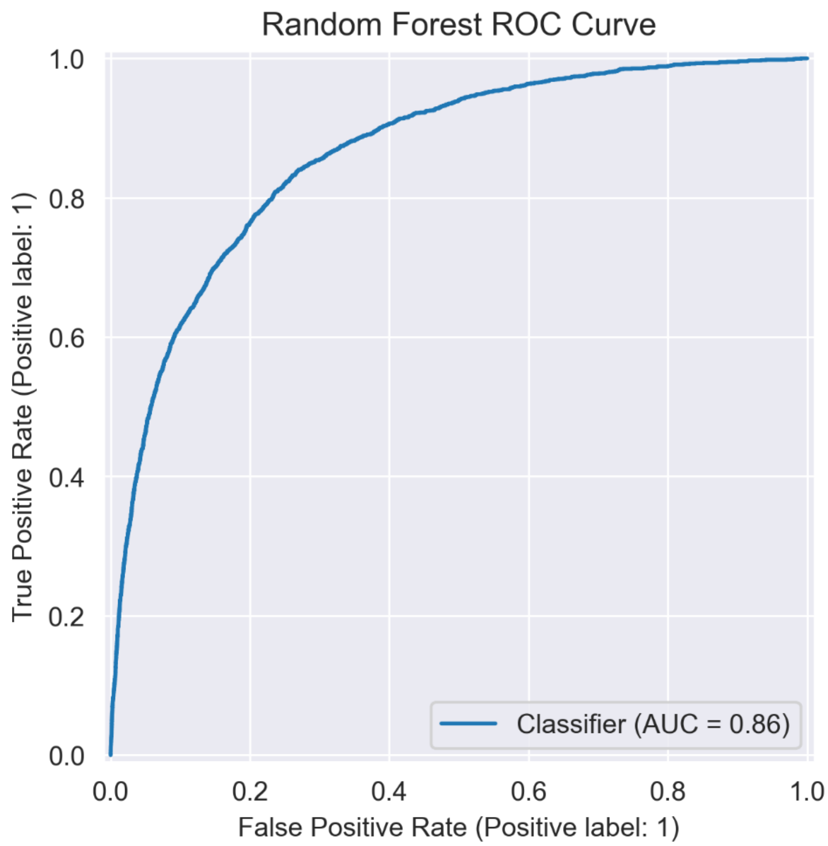

Figure 4. Random Forest ROC curve.

Figure 4 shows the ROC curve for the Random Forest model. The ROC-AUC score of approximately 0.864 indicates that the model has strong ability to distinguish between default and non-default borrowers.

Misclassified cases were also examined. Several misclassified borrowers showed mixed financial signals, such as high credit utilization, moderate debt ratios, previous late payment behavior, or missing income values. These cases are difficult because borrower risk is not always clearly separable. This highlights a key limitation of financial prediction systems: real-world borrower behavior is complex, and no model can perfectly classify every case.

10. Interpretability Analysis

Model interpretability is important in financial machine learning because credit-related predictions can affect real individuals. A model should not only produce predictions but also provide understandable explanations of which features contribute to risk.

10.1 Random Forest Feature Importance

Random Forest feature importance was used to identify the most influential predictors in the model.

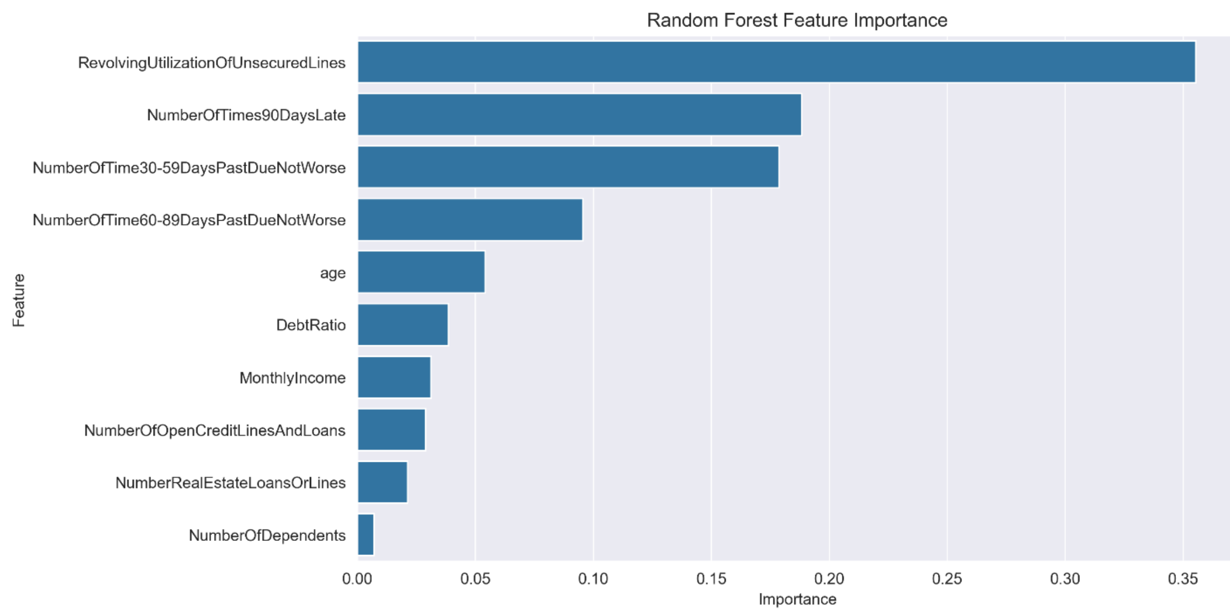

Figure 5. Random Forest feature importance.

Figure 5 shows that the most important feature was `RevolvingUtilizationOfUnsecuredLines`. This feature represents how much of a borrower's available unsecured credit is being used. High utilization often indicates financial stress and was strongly associated with default risk.

Other important features included:

- `NumberOfTimes90DaysLate`,
- `NumberOfTime30-59DaysPastDueNotWorse`,
- `NumberOfTime60-89DaysPastDueNotWorse`,
- `age`,
- `DebtRatio`.

These results are financially meaningful. Borrowers with repeated late payment history are more likely to experience future default. This confirms that delinquency history is one of the strongest indicators of credit risk.

10.2 SHAP Analysis

SHAP analysis was used to provide a more detailed explanation of feature effects. Unlike standard feature importance, SHAP shows both the importance and the direction of feature influence on predictions (Lundberg & Lee, 2017).

Figure 6. SHAP summary plot.

Figure 6 shows the SHAP summary plot. The plot confirms that high revolving credit utilization strongly increases predicted default risk. Delinquency-related variables also strongly push predictions toward higher default probability. Age appears to have a directional relationship, where younger borrowers tend to receive slightly higher risk predictions in many cases. Income and debt-related variables contribute moderately compared with utilization and payment history.

The SHAP analysis supports the earlier Random Forest feature importance results while providing more detailed insight into how feature values affect model output.

11. Risk Scoring System

A risk scoring system was developed using the predicted default probabilities from the Random Forest model. Instead of only predicting default or non-default, the model produces a probability that can be converted into practical risk categories.

Borrowers were classified into three risk bands:

Predicted Default Probability	Risk Band
Less than 0.20	Low Risk
0.20 to 0.50	Medium Risk
Greater than or equal to 0.50	High Risk

This approach makes the model more useful for business decision-making. A bank or lending institution could use these categories to support loan review processes, adjust interest rates, request additional documentation, or apply closer manual review to high-risk borrowers.

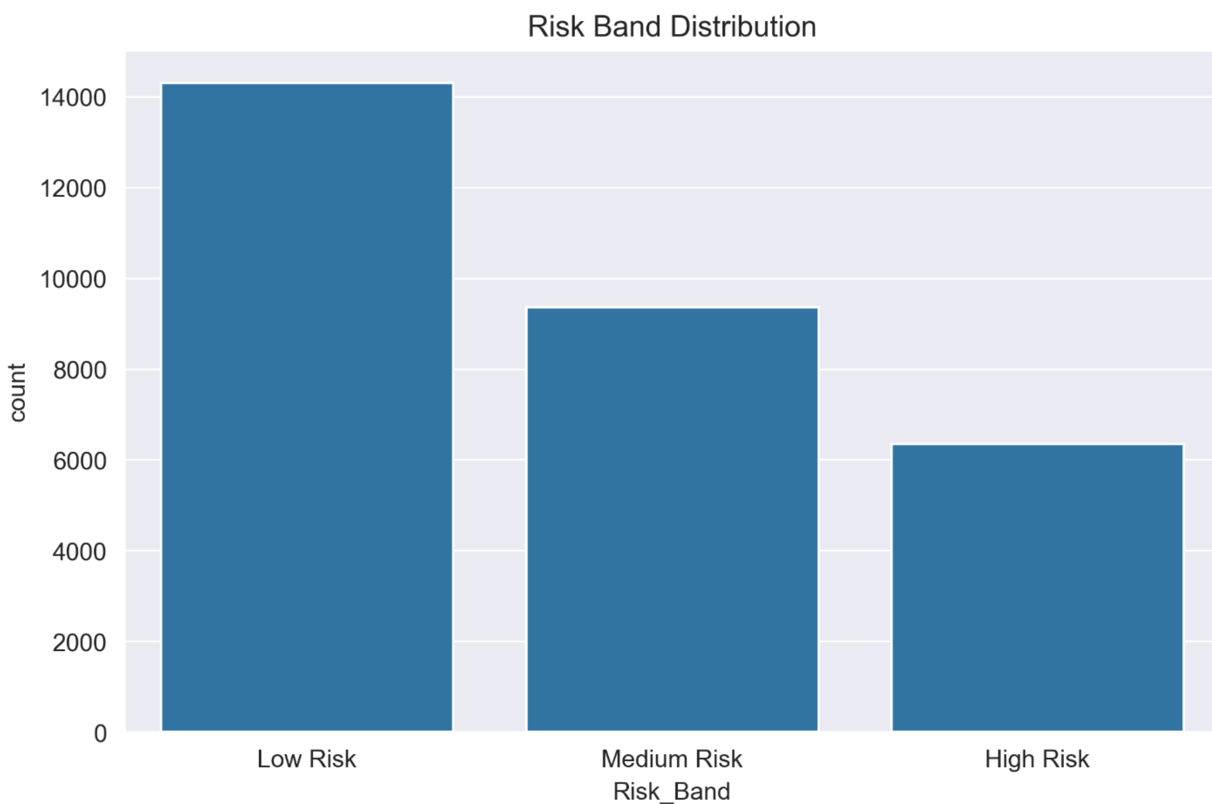

Figure 7. Risk band distribution.

Figure 7 shows the distribution of risk bands in the test set. Most borrowers were classified as Low Risk, while smaller groups were classified as Medium Risk or High Risk. This reflects the original class imbalance in the dataset, where most borrowers did not default.

The risk scoring system demonstrates how machine learning predictions can be translated into business-friendly categories. However, these categories should support human decision-making rather than replace it.

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12. Fairness, Ethics, and Limitations

Machine learning models in lending systems raise important fairness and ethical concerns. Credit-related decisions can affect access to loans, housing, education, business opportunities, and financial stability. Therefore, predictive systems must be used responsibly.

One concern is the use of features such as age and income. Although these variables may improve predictive performance, they can also create fairness concerns. Age may directly relate to protected demographic characteristics, while income can act as a proxy for socioeconomic status. If a model learns patterns from historically biased data, it may reproduce or amplify unfair outcomes.

Another concern is interpretability. Financial institutions should be able to explain why a borrower is classified as risky. Black-box models may create trust and accountability problems, especially if borrowers are denied credit without understandable reasons. This project addressed interpretability using feature importance and SHAP analysis, but further fairness testing would be needed before real-world deployment.

The dataset also has limitations. It contains missing values, outliers, and class imbalance. In addition, the data may not fully represent all borrower populations or current lending environments. Model performance on this dataset does not guarantee equal performance in real-world financial systems.

The model should therefore be treated as a decision-support tool, not as a fully automated decision-maker. Human analysts should review high-impact lending decisions, especially borderline cases or cases involving unusual borrower profiles.

Key limitations of this project include:

- missing values in important features,
- extreme outliers in financial variables,
- strong class imbalance,
- possible historical bias in the data,
- limited fairness evaluation,
- and lack of real-world deployment testing.

Future work should include fairness metrics, threshold optimization, external validation, and monitoring of model performance over time.

13. Conclusion

This project developed a machine learning workflow for predicting loan default risk using the “Give Me Some Credit” dataset. The project included exploratory data analysis, preprocessing, baseline modeling, advanced modeling, hyperparameter tuning, model evaluation, error analysis, interpretability analysis, and risk scoring.

The results showed that accuracy alone was not sufficient for evaluating model performance because the dataset was highly imbalanced. The baseline Logistic Regression model achieved high accuracy but failed to detect most default cases. Balanced Logistic Regression improved recall, but with lower precision and accuracy.

The Random Forest models achieved the strongest overall performance. Both the original and tuned Random Forest models reached ROC-AUC values of approximately 0.864, showing strong ability to distinguish between default and non-default borrowers. Hyperparameter tuning demonstrated the trade-off between precision and recall, while SHAP analysis confirmed that revolving credit utilization and delinquency history were the most important drivers of predicted default risk.

The risk scoring system converted predicted probabilities into Low Risk, Medium Risk, and High Risk categories. This made the model outputs more understandable and practical for financial decision-support use cases.

Overall, the project demonstrates that machine learning can support loan default prediction and financial risk assessment. However, the model should be used responsibly, with attention to fairness, transparency, and human oversight. Future improvements may include additional ensemble models, advanced imbalance handling, threshold optimization, expanded fairness analysis, and deployment as a web application or decision-support tool.

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Appendix: Repository Contents

The final project repository includes:

- `notebooks/01_EDA.ipynb`: full machine learning workflow notebook,
- `figures/`: saved visualizations,
- `results/`: exported model comparison and feature importance tables,
- `models/`: saved trained Random Forest model,
- `reports/`: model card, executive summary, and final report,
- `README.md`: project overview and usage instructions,
- `requirements.txt`: required Python libraries.
